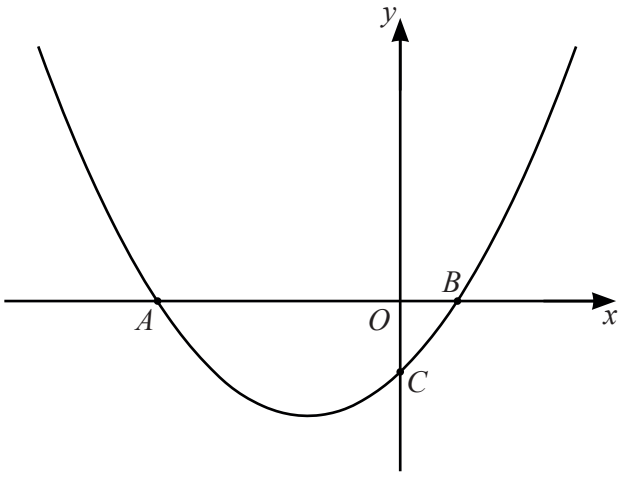
0580_w20_qp_41_Q10

10 (a)



NOT TO SCALE

The diagram shows a sketch of the curve $y = x^2 + 3x - 4$.

(i) Find the coordinates of the points A, B and C.

A (..... , ...)

B (..... , ...)

C (..... , ...) [4]

(ii) Differentiate $x^2 + 3x - 4$.

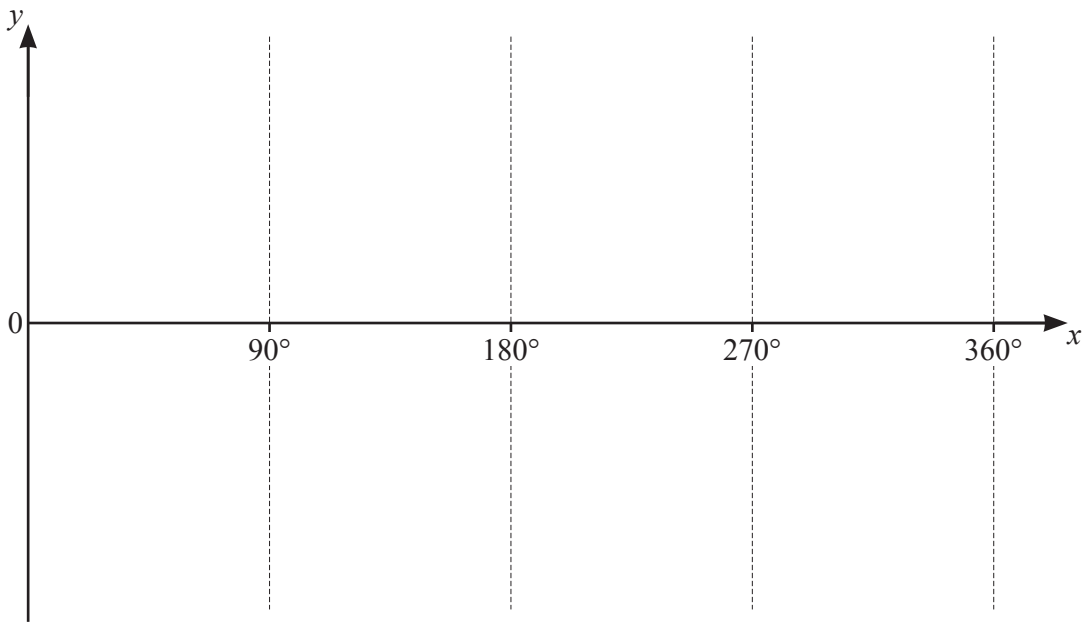
..... [2]

(iii) Find the equation of the tangent to the curve at the point (2, 6).

..... [3]

0580_w20_qp_41_Q10

(b)



(i) On the diagram, sketch the graph of $y = \tan x$ for $0° \leqslant x \leqslant 360°$. [2]

(ii) Solve the equation $5 \tan x = -7$ for $0° \leqslant x \leqslant 360°$.

$x =$ ..... or $x =$ ..... [3]